

To:

The Editor-in-Chief
Neurocomputing
Elsevier B.V.

Date: September 5, 2026

Subject: Submission of original research manuscript titled “*Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models*”

Dear Editor-in-Chief,

I am pleased to submit the above-titled original research paper for consideration for publication in *Neurocomputing*.

The manuscript develops a common output-space read abstraction for finite attention, external retrieval, and parameter-expert routing. In particular, it introduces a measure-theoretic formulation with explicit existence conditions and support for query-dependent expert outputs, establishes exact ranked top- k truncation bounds alongside an error-budgeted adaptive rule and an approximate signed-cancellation bound, and presents controlled synthetic and CounterFact studies investigating external memory, approximate nearest-neighbor candidates, multi-space routing, signed cancellation, and GPT-2 factual steering.

The experiments are intentionally presented as mechanism studies. Trainable synthetic experiments use five random seeds, while the manuscript explicitly details operational limitations, including the absence of large-scale approximate nearest-neighbor evaluations, competitive end-to-end model-editing baselines, and certified unlearning guarantees. The complete codebase, benchmark definitions, result files, and reproduction scripts are publicly accessible at <https://github.com/mossaiby/RoAS>.

I confirm that the manuscript is original and is not under consideration elsewhere, the author declares no competing financial or personal interests, and the use of generative AI during manuscript preparation is disclosed in the manuscript in full accordance with Elsevier policy.

The paper may be relevant to readers interested in neural memory, retrieval, sparse routing, and mathematical descriptions of attention. Thank you for considering the manuscript.

Sincerely,

Farshid Mossaiby, Ph.D.

Associate Professor

Department of Civil Engineering

University of Isfahan, Isfahan, 81744-73441, Iran

Email: mossaiby@eng.ui.ac.ir

GitHub Repository: <https://github.com/mossaiby/RoAS>